Text Preprocessing - 2

* TF - IDF :- [Term Frequency - Inverse Document Frequency]

- A type of technique that is used to encode the text data.
- It is an improvement to count BOW.
- It counts the occurrence of words along with measuring the importance of a word in a document / sentence.
- It up-weights the rare words.

Term Frequency :-

↳ It calculates how often a word / term appears in a document.

$$TF = \frac{\text{No. of times a term appears in a document}}{\text{Total terms in a document}}$$

Inverse - Document Frequency :-

↳ It calculates the rareness of a word / term across all documents.

$$\text{Document Frequency (DF)} = \frac{\text{No. of documents containing a term}}{\text{Total no. of documents}}$$

Ex- Let us take two words →

w_1 = pizza $\qquad$ w_2 = tasty

if total documents = 10 & documents containing pizza = 7

" " tasty = 2

∴ total documents = 10 & documents containing "pizza" = 7
" " " "tasty" = 2

then, $DF_{(pizza)} = \frac{7}{10} = 0.7$; $DF_{(tasty)} = \frac{2}{10} = 0.2$

DF gives a high weightage to pizza & low weightage to tasty.

∴ On inverting,

$$IDF = \log\left[\frac{\text{Total no. of documents}}{\text{No. of documents containing a term}}\right]$$

$IDF_{(pizza)} = \log\left(\frac{10}{7}\right)$; $IDF_{(tasty)} = \log\left(\frac{10}{2}\right)$

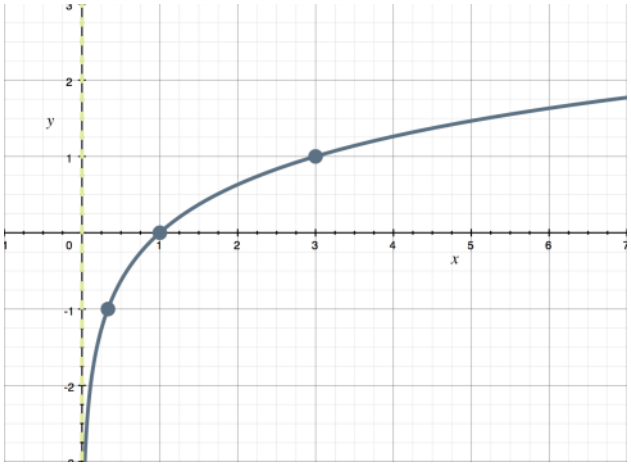
$= \log(1.4)$; $= \log(5)$

$= 0.14$; $= 0.69$

∴ tasty → high weightage ⇝ It is rare
pizza → low weightage ⇝ It is common

Why log?

↳ To dampen the term with high frequency.



$\log(1) = 0$
$\log(10) = 1$
$\log(100) = 2$
$\log(1000) = 3$
⋮

Suppose → A word w_1 occurs 100k times in a document.
A word w_1 occurs 200k times in a document.

We observe no much difference in the relevancy as the word is important for both the documents.

∴ to dampen the frequency we use log & calculate IDF

After calculating TF & IDF separately, we combine TF - IDF

TF-IDF = TF * IDF

* Drawbacks of TF-IDF :-
  - Ignores the word order. ⇝ dog bites man = man bites dog
  - High dimensional data.
  - Problem of sparsity ⇝ more no. of zeroes
  - Sensitive to new term
  - No semantic relationship
    - ↳ Word → Bank

River Bank

Bank Account

↳ meaning of the word bank is different in both phrases ↲